

Knowledge Representation

Lecture 2:

Actions with Ramifications

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Basic assumptions

- Inertia law.
- Complete information about all actions and all fluents.
- Nondeterminism is allowed.
- Side effects (ramifications) of actions are admitted.
- Action can be unexecutable in some states.
- If an action precondition does not hold, the execution of the action has no effect.

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To represent this class of dynamic systems we will use action languages of the class *AR*.

Action Language $\mathcal{AR}$ – Syntax

$\mathcal{AR}$ is a family of languages, each one of which is characterised by a signature $\Upsilon = (\mathcal{F}, \mathcal{Ac})$. Moreover, $\mathcal{F}_I \subseteq \mathcal{F}$.

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Formula – any propositional combination of fluents:

$$\alpha ::= f \mid \neg\alpha \mid \alpha_1 \wedge \alpha_2 \mid \alpha_1 \vee \alpha_2 \mid \alpha_1 \rightarrow \alpha_2 \mid \alpha_1 \leftrightarrow \alpha_2.$$

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Two specific formulas:

$\top$:	truth
$\perp$:	falsity.

Statements in $\mathcal{AR}$

• *Value statement:*

α *after* $A_1, \dots, A_n$

observable α *after* $A_1, \dots, A_n$

where α is a formula and $A_i \in \mathcal{Ac}$, $i = 1, \dots, n$, are actions.

These statements are intuitively read as:

- The condition α *always* (must) holds after performing the sequence $A_1, \dots, A_n$ of actions.
- The condition α *sometimes* (may) holds after performing the sequence $A_1, \dots, A_n$ of actions.

If $n = 0$ then we use the abbreviation *initially* α

Statements in $\mathcal{AR}$ (cont.)

• *Effect statement:*

A *causes* α *if* π

where $\alpha, \pi \in \text{Forms}(\mathcal{F})$ and $A \in \mathcal{A}_c$.

Intuitively:

Performing the action A in any state satisfying π leads to the effect α .

Statements in $\mathcal{AR}$ (cont.)

• *Effect statement:*

A causes α if π

where $\alpha, \pi \in \text{Forms}(\mathcal{F})$ and $A \in \mathcal{A}_c$.

Intuitively:

Performing the action A in any state satisfying π leads to the effect α .

Abbreviations:

(i) $\pi \Leftrightarrow \top$: *A causes α*

(ii) $\alpha \Leftrightarrow \perp$: *impossible A if π*

(ii) says that the action A is unexecutable in any state, where π holds.

Statements in $\mathcal{AR}$ (cont.)

• *Release statement:*

A releases f if π

where $A \in \mathcal{A}_c$, $f \in \mathcal{F}_I$, and $\pi \in \text{Forms}(\mathcal{F})$.

Intuitively:

*Performing the action A in any state satisfying π **might**, but **need not**, change the value of the inertial fluent f .*

Statements in $\mathcal{AR}$ (cont.)

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*Performing the action A in any state satisfying π **might**, but **need not**, change the value of the inertial fluent f .*

Abbreviation:

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E.g., Toss **releases** heads.

Statements in $\mathcal{AR}$ (cont.)

• *Constraint statement:*

always α

where $\alpha \in \text{Forms}(\mathcal{F})$.

Intuitively:

Every state satisfies the condition α .

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Intuitively:

Every state satisfies the condition α .

E.g., *always walking $\rightarrow$ alive.*

Statements in $\mathcal{AR}$ (cont.)

- *Fluent specification statement:*

noninertial f

where $f \notin \mathcal{F}_I$.

Intuitively, it specifies the fluent f as *noninertial*.

An *action domain* is a finite set of statements.

Example: Russian Turkey Scenario (RTS)

There is a shooter Bill and a turkey Fred. Bill can load the gun, which makes it loaded. Bill can also spin the gun, which randomly makes it unloaded. Shooting the gun makes it unloaded and, if the gun was loaded, makes Fred dead. Initially the gun was unloaded and Fred was alive.

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Representation in *AR*:

***initially** loaded $\wedge$ alive;*

*LOAD **causes** loaded;*

*SPIN **releases** loaded;*

*SHOOT **causes** $\neg$ loaded;*

*SHOOT **causes** $\neg$ alive **if** loaded;*

Example: Opening the door

Consider a scenario:

Inserting a card makes the door open. If one has no card, it is not possible to insert it. Initially the door are closed.

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Representation in *AR*:

initially $\neg open$;

INSERTCARD ***causes*** open;

impossible INSERTCARD ***if*** $\neg hasCard$.

Example: Two switches

Consider a scenario:

*There are two switches, which can be in the position on or off. If both switches are in the same position, the light is on, otherwise it is off.
Pressing a switch makes it changed its position.*

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Pressing a switch makes it changed its position.*

Representation in *AR*:

noninertial light;
always light $\equiv$ (*switch1* $\equiv$ *switch2*);
TOGGLE1 **causes** *switch1* **if** $\neg$ *switch1*;
TOGGLE1 **causes** $\neg$ *switch1* **if** *switch1*;
TOGGLE2 **causes** *switch1* **if** $\neg$ *switch2*;
TOGGLE2 **causes** $\neg$ *switch2* **if** *switch2*.

Semantics of $\mathcal{AR}$

A **state** is a mapping $\sigma : \mathcal{F} \rightarrow \{0, 1\}$

It is extended to the mapping σ^* defined for all formulas over $\mathcal{F}$ according to well-known rules in classical propositional logic:

$$\sigma^*(f) = \sigma(f) \text{ for } f \in \mathcal{F}$$

$$\sigma^*(\neg\alpha) = 1 - \sigma^*(\alpha)$$

$$\sigma^*(\alpha \wedge \beta) = \min(\sigma^*(\alpha), \sigma^*(\beta))$$

$$\sigma^*(\alpha \vee \beta) = \max(\sigma^*(\alpha), \sigma^*(\beta))$$

$$\sigma^*(\alpha \rightarrow \beta) = \sigma^*(\neg\alpha \vee \beta)$$

$$\sigma^*(\alpha \equiv \beta) = \sigma^*((\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)).$$

For any formula $\alpha \in \text{Forms}(\mathcal{F})$, if $\sigma^*(\alpha) = 1$, then we write $\sigma \models \alpha$

Semantics of $\mathcal{AR}$ – structures

Definition 2.1 A *structure for a language $\mathcal{L}$ of the class $\mathcal{AR}$* is a tuple $S = (\Sigma, \sigma_0, Res)$ where

- Σ is a non-empty set of states
- $\sigma_0 \in \Sigma$ is an initial state
- $Res : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is a partial mapping called a transition function.

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Intuitively, for any action $A \in \mathcal{Ac}$ and for any state $\sigma \in \Sigma$, Res assigns all states accessible by performing A in a state σ .

Moreover,

- $Res(A, \sigma)$ are all such states, which are as “close” to σ as possible
- Σ is the set of states satisfying all constraints.

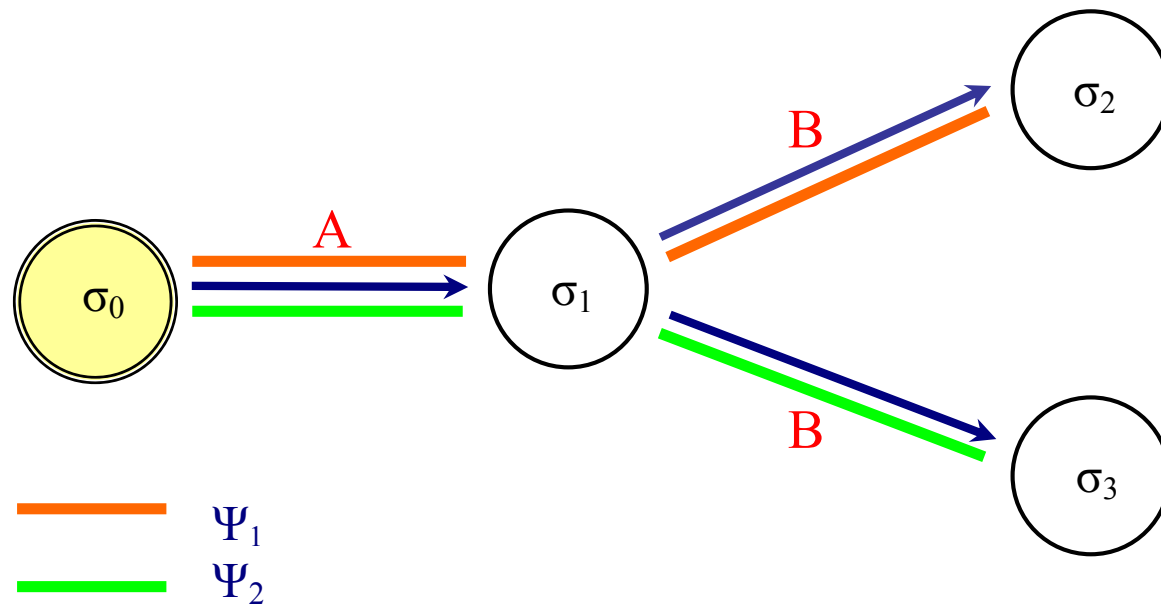
Semantics of $\mathcal{AR}$ (cont.)

Let D be an action domain.

Given a structure $S = (\Sigma, \sigma_0, Res)$, define a partial mapping $\Psi_S : \mathcal{Ac}^* \times \Sigma \rightarrow \Sigma$ as follows:

- $\Psi_S(\varepsilon, \sigma) = \sigma$
- if for $n \geq 1$, $\Psi_S((A_1, \dots, A_n), \sigma)$ is defined and $Res((A_1, \dots, A_n, A_{n+1}), \sigma) \neq \emptyset$, then $\Psi_S((A_1, \dots, A_n, A_{n+1}), \sigma) \in Res(A_n, \Psi_S((A_1, \dots, A_n), \sigma))$.

Mapping Ψ depicted



Semantics of $\mathcal{AR}$ (cont.)

Let D be an action domain and let $S = (\Sigma, \sigma_0, Res)$ be a structure for $\mathcal{AR}$.

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We say that

- a fluent $f \in \mathcal{F}$ is **inertial in D** iff **(noninertial f)** $\notin D$

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- a fluent $f \in \mathcal{F}$ is **inertial in D** iff $(\mathbf{noninertial} \ f) \notin D$
- $\sigma : \mathcal{F} \rightarrow \{0, 1\}$ is a **state of D** iff for every constraint $(\mathbf{always} \ \alpha) \in D$, it holds $\sigma \models \alpha$

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- a value statement **(α after $A_1, \dots, A_n$)** is **true** in S iff $\Psi_S((A_1, \dots, A_n), \sigma_0) \models \alpha$ for **every** Ψ_S defined as before

Semantics of $\mathcal{AR}$ (cont.)

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- $\sigma : \mathcal{F} \rightarrow \{0, 1\}$ is a **state of D** iff for every constraint (**always** α) $\in D$, it holds $\sigma \models \alpha$
- a value statement (α **after** $A_1, \dots, A_n$) is **true** in S iff $\Psi_S((A_1, \dots, A_n), \sigma_0) \models \alpha$ for **every** Ψ_S defined as before
- an observation statement (**observable** α **after** $A_1, \dots, A_n$) is **true in D** iff $\Psi_S((A_1, \dots, A_n), \sigma_0) \models \alpha$ for **some** Ψ_S defined as before.

Semantics of $\mathcal{AR}$ (cont.)

Let D be an action domain and let $S = (\Sigma, \sigma_0, Res)$ and a structure for $\mathcal{AR}$.

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Let D be an action domain and let $S = (\Sigma, \sigma_0, Res)$ and a structure for $\mathcal{AR}$.

First, define the following auxiliary mapping $Res_0 : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ as follows:
for every $A \in \mathcal{Ac}$ and for every $\sigma \in \Sigma$,

$$Res_0(A, \sigma) = \{\sigma' \in \Sigma : (A \textbf{ causes } \alpha \textbf{ if } \pi) \in D \ \& \ \sigma \models \pi \Rightarrow \sigma' \models \alpha\}$$

Semantics of $\mathcal{AR}$ (cont.)

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Furthermore, for all $\sigma, \sigma' \in \Sigma$ and for every $A \in \mathcal{Ac}$, let $New(A, \sigma, \sigma')$ be the set of all literals $\bar{f}$ such that $\sigma' \models \bar{f}$ and

- f is inertial ($f \in \mathcal{F}_I$) and $\sigma(f) \neq \sigma'(f)$, or
- for some statement $(A \textbf{ releases } f \textbf{ if } \pi) \in D$ it holds $\sigma \models \pi$.

Semantics of $\mathcal{AR}$ – model

Definition 2.2 Let D be an action domain in $\mathcal{AR}$ and let $S = (\Sigma, \sigma_0, Res)$ be a structure for $\mathcal{AR}$. We say that S is the *model of D* iff

- (M.1) Σ is the set of all states for D
- (M.2) every value statement and every observation statement is true in S
- (M.3) for any $A \in \mathcal{Ac}$ and for any $\sigma \in \Sigma$, $Res(A, \sigma)$ is the set of all states $\sigma' \in \Sigma$ for which the sets $New(A, \sigma, \sigma')$ are minimal (wrt set inclusion).

Example: YSP (cond.)

Recall the YSP:

initially loaded $\wedge$ walking;

always walking $\rightarrow$ alive;

LOAD *causes* loaded;

SHOOT *causes* $\neg$ loaded;

SHOOT *causes* $\neg$ alive *if* loaded.

Here we have $\Sigma = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5\}$, where

$$\sigma_0 = \{a, \neg l, w\}$$

$$\sigma_3 = \{a, l, w\}$$

$$\sigma_1 = \{a, \neg l, \neg w\}$$

$$\sigma_4 = \{a, l, \neg w\}$$

$$\sigma_2 = \{\neg a, \neg l, \neg w\}$$

$$\sigma_5 = \{\neg a, l, \neg w\}.$$

Example: YSP (cont.)

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

$$Res_0(\text{LOAD}, \sigma_0) = Res_0(\text{LOAD}, \sigma_1) = Res_0(\text{LOAD}, \sigma_2) = \{\sigma_3, \sigma_4, \sigma_5\}.$$

$$\begin{array}{l} * \quad New(\text{LOAD}, \sigma_0, \sigma_3) = \{l\} \\ \quad \quad New(\text{LOAD}, \sigma_0, \sigma_4) = \{l, \neg w\} \\ \quad \quad New(\text{LOAD}, \sigma_0, \sigma_5) = \{\neg a, l, \neg w\} \\ \quad \quad \underline{Res(\text{LOAD}, \sigma_0) = \{\sigma_3\}.} \end{array}$$

Example: YSP (cont.)

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

$$Res_0(\text{LOAD}, \sigma_3) = \{\sigma_3, \sigma_4, \sigma_5\}$$

$$* \quad New(\text{LOAD}, \sigma_3, \sigma_3) = \emptyset$$

$$New(\text{LOAD}, \sigma_3, \sigma_4) = \{\neg w\}$$

$$New(\text{LOAD}, \sigma_3, \sigma_5) = \{\neg a, \neg w\}$$

$$\underline{Res(\text{LOAD}, \sigma_3) = \{\sigma_3\}}.$$

Example: YSP (cont.)

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

Similarly,

$$Res_0(\text{LOAD}, \sigma_4) = Res_0(\text{LOAD}, \sigma_5) = \{\sigma_3, \sigma_4, \sigma_5\},$$

therefore,

$$\begin{array}{l}Res(\text{LOAD}, \sigma_4) = \{\sigma_4\} \\ Res(\text{LOAD}, \sigma_5) = \{\sigma_5\}.\end{array}$$

Example: YSP (cont.)

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

$$Res_0(\text{SHOOT}, \sigma_0) = Res_0(\text{SHOOT}, \sigma_1) = Res_0(\text{SHOOT}, \sigma_2) = \{\sigma_0, \sigma_1, \sigma_2\}.$$

$$\begin{array}{l} * \quad New(\text{SHOOT}, \sigma_0, \sigma_0) = \emptyset \\ \quad \quad New(\text{SHOOT}, \sigma_0, \sigma_1) = \{\neg w\} \\ \quad \quad New(\text{SHOOT}, \sigma_0, \sigma_2) = \{\neg a, \neg w\} \\ \quad \quad \underline{Res(\text{SHOOT}, \sigma_0) = \{\sigma_0\}.} \end{array}$$

Example: YSP (cont.)

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Analogously we calculate

$$\begin{array}{l} Res(\text{SHOOT}, \sigma_1) = \{\sigma_1\} \\ Res(\text{SHOOT}, \sigma_2) = \{\sigma_2\}. \end{array}$$

Example: YSP (cont.)

$$\begin{array}{ll}\sigma_0 = \{a, \neg l, w\} & \sigma_3 = \{a, l, w\} \\ \sigma_1 = \{a, \neg l, \neg w\} & \sigma_4 = \{a, l, \neg w\} \\ \sigma_2 = \{\neg a, \neg l, \neg w\} & \sigma_5 = \{\neg a, l, \neg w\}.\end{array}$$

Note: $Res_0(\text{SHOOT}, \sigma_3) = \{\sigma_2\}$.

Immediately we get $Res(\text{SHOOT}, \sigma_3) = \{\sigma_2\}$.

Example: YSP (cont.)

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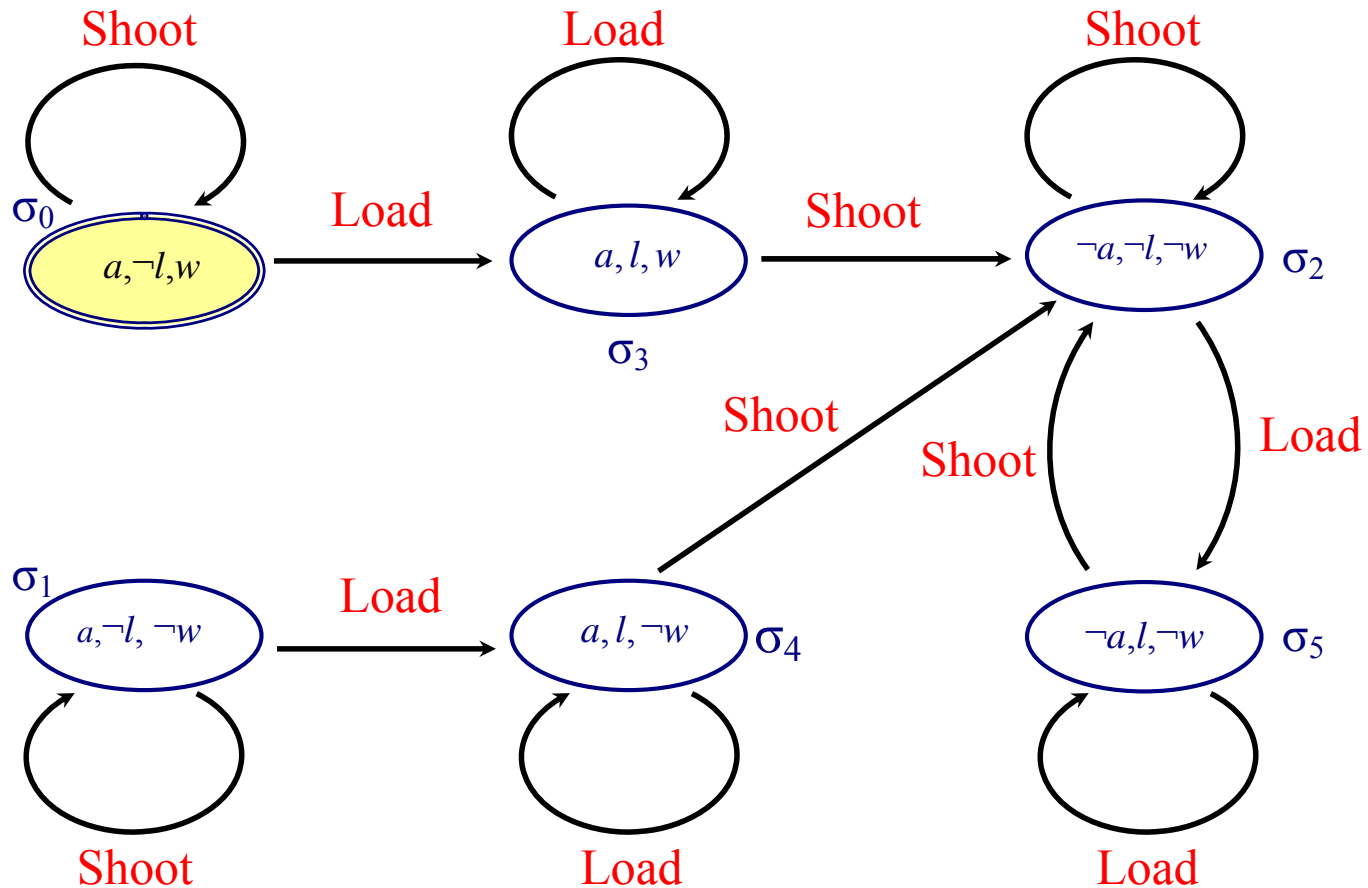
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so

$$Res(\text{SHOOT}, \sigma_4) = Res(\text{SHOOT}, \sigma_5) = \{\sigma_2\}.$$

Example: YSP (cont.)



Example: Two switches

Recall:

*There are two switches, which can be in the position on or off. If both switches are in the same position, the light is on, otherwise it is off.
Pressing any switch changes its position.*

Representation in *AR*:

noninertial *light*;
initially $switch_1 \wedge switch_2$;
always $light \equiv (switch_1 \equiv switch_2)$;
TOGGLE1 ***causes*** $\neg switch_1$ ***if*** $switch_1$;
TOGGLE1 ***causes*** $switch_1$ ***if*** $\neg switch_1$;
TOGGLE2 ***causes*** $\neg switch_2$ ***if*** $switch_2$;
TOGGLE2 ***causes*** $switch_2$ ***if*** $\neg switch_2$.

Example: Two switches (cont.)

Here we have $\Sigma = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$, where

$$\begin{aligned}\sigma_0 &= \{switch_1, switch_2, light\} & \sigma_2 &= \{switch_1, \neg switch_2, \neg light\} \\ \sigma_1 &= \{\neg switch_1, \neg switch_2, light\} & \sigma_3 &= \{\neg switch_1, switch_2, \neg light\}.\end{aligned}$$

Assume that all fluents are inertial. Then we have:

$$Res_0(\text{TOGGLE1}, \sigma_0) = \{\sigma_1, \sigma_3\}$$

$$* \quad New(\text{TOGGLE1}, \sigma_0, \sigma_1) = \{\neg switch_1, \neg switch_2\}$$

$$* \quad New(\text{TOGGLE1}, \sigma_0, \sigma_3) = \{\neg \sigma_1, \neg light\}$$

$$\underline{Res(\text{TOGGLE1}, \sigma_0) = \{\sigma_1, \sigma_3\}.$$

So TOGGLE1 in σ_0 is non-deterministic, but the resulting state σ_1 is counterintuitive!

Example: Two switches (cont.)

$$\sigma_0 = \{switch_1, switch_2, light\}$$

$$\sigma_2 = \{switch_1, \neg switch_2, \neg light\}$$

$$\sigma_1 = \{\neg switch_1, \neg switch_2, light\}$$

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Similarly,

$$Res_0(\text{TOGGLE1}, \sigma_1) = \{\sigma_0, \sigma_2\}$$

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$$\underline{Res(\text{TOGGLE1}, \sigma_1) = \{\sigma_0, \sigma_2\}.$$

Note that only the resulting state σ_2 coincides with our intuition.

Example: Two switches (cont.)

$$\begin{aligned}\sigma_0 &= \{switch_1, switch_2, light\} & \sigma_2 &= \{switch_1, \neg switch_2, \neg light\} \\ \sigma_1 &= \{\neg switch_1, \neg switch_2, light\} & \sigma_3 &= \{\neg switch_1, switch_2, \neg light\}.\end{aligned}$$

Now let the fluent *light* be *noninertial*. Then

$$\begin{aligned}Res_0(\text{TOGGLE1}, \sigma_0) &= \{\sigma_1, \sigma_3\} \\ New(\text{TOGGLE1}, \sigma_0, \sigma_1) &= \{\neg switch_1, \neg switch_2\} \\ * \quad New(\text{TOGGLE1}, \sigma_0, \sigma_3) &= \{\neg switch_1\} \\ \underline{Res(\text{TOGGLE1}, \sigma_0)} &= \{\sigma_3\}.\end{aligned}$$

Although $\sigma_0(light) \neq \sigma_3(light)$, we have $light \notin \mathcal{F}_I$ (noninertial), so $\neg light \notin New(\text{TOGGLE1}, \sigma_0, \sigma_3)$.

Example: Two switches (cont.)

$$\sigma_0 = \{switch_1, switch_2, light\}$$

$$\sigma_1 = \{\neg switch_1, \neg switch_2, light\}$$

$$\sigma_2 = \{switch_1, \neg switch_2, \neg light\}$$

$$\sigma_3 = \{\neg switch_1, switch_2, \neg light\}.$$

Also,

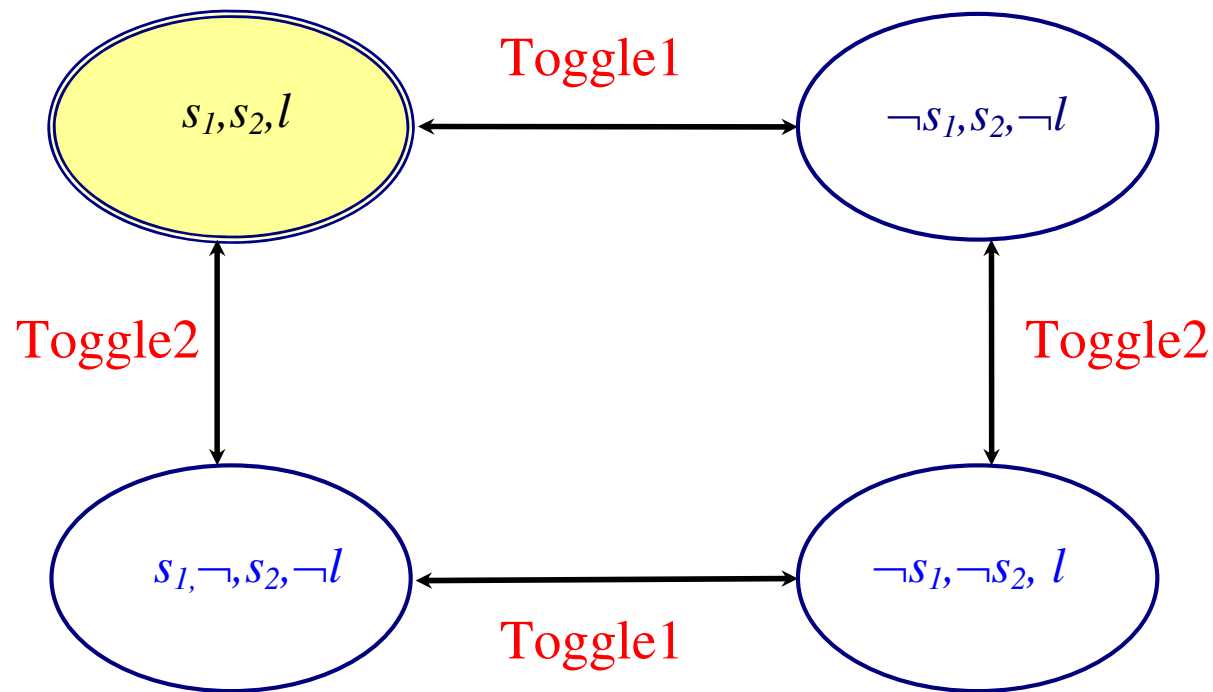
$$Res_0(\text{TOGGLE1}, \sigma_1) = \{\sigma_0, \sigma_2\}$$

$$New(\text{TOGGLE1}, \sigma_1, \sigma_0) = \{switch_1, switch_2\}$$

$$* \quad New(, \sigma_1, \sigma_2) = \{switch_1\}$$

$$\underline{Res(\text{TOGGLE1}, \sigma_1) = \{\sigma_2\}.$$

Example: Two switches (cont.)



Example: Buying a paper

Consider an action **BUY** which causes that an agent has a paper A (*hasA*)
or a paper B (*hasB*).

BUY *causes* $hasA \vee hasB$.

Put

$$\sigma_1 = \{\neg hasA, \neg hasB\}$$

$$\sigma_3 = \{hasA, \neg hasB\}$$

$$\sigma_2 = \{\neg hasA, hasB\}$$

$$\sigma_4 = \{hasA, hasB\}.$$

Then

$$Res_0(\mathbf{BUY}, \sigma_1) = \{\sigma_2, \sigma_3, \sigma_4\}$$

$$* \quad New(\mathbf{BUY}, \sigma_1, \sigma_2) = \{hasB\}$$

$$* \quad New(\mathbf{BUY}, \sigma_1, \sigma_3) = \{hasA\}$$

$$New(\mathbf{BUY}, \sigma_1, \sigma_4) = \{hasA, hasB\}$$

$$Res(\mathbf{BUY}, \sigma_1) = \{\sigma_2, \sigma_3\}$$

Example: Buying a paper (cont.)

Problem:

How to obtain the effect of this action such that an agent has both papers?

Example: Buying a paper (cont.)

Problem:

How to obtain the effect of this action such that an agent has both papers?

A possible solution:

BUY **causes** $hasA \vee hasB$;

BUY **releases** $hasA$ **if** $\neg hasA$;

BUY **releases** $hasB$ **if** $\neg hasB$;

Example: Buying a paper (cont.)

$$\sigma_1 = \{\neg hasA, \neg hasB\}$$

$$\sigma_2 = \{\neg hasA, hasB\}$$

$$\sigma_3 = \{hasA, \neg hasB\}$$

$$\sigma_4 = \{hasA, hasB\}.$$

So we get:

$$Res_0(\text{BUY}, \sigma_1) = \{\sigma_2, \sigma_3, \sigma_4\}$$

$$* \quad New(\text{BUY}, \sigma_1, \sigma_2) = \{\neg hasA, hasB\}$$

$$* \quad New(\text{BUY}, \sigma_1, \sigma_3) = \{hasA, \neg hasB\}$$

$$* \quad New(\text{BUY}, \sigma_1, \sigma_4) = \{hasA, hasB\}$$

$$\underline{Res(\text{BUY}, \sigma_1) = \{\sigma_2, \sigma_3, \sigma_4\}.$$

Thank you for your attention!

Any questions are welcome.